

LESSON 5-8 • SESSION 1 • GUIDED NOTES

Surface Area of Pyramids

Learning target – I can find the surface area of a pyramid by adding the base area and the areas of its triangular faces.
6.G.A.4

WORDS TO KNOW

fill in the missing words

WORD	WHAT IT MEANS	EXAMPLE
slant height <i>altura inclinada</i>	The height of a side triangle, measured _____ its face.	$\frac{1}{2} \times \text{base} \times \text{slant height}$
lateral face <i>cara lateral</i>	A side _____ of a pyramid.	4 on a square pyramid
apex <i>ápice</i>	The _____ point where the side faces meet.	tip of the pyramid

KEY IDEA

complete each sentence

- SA of a pyramid = base area + area of all the _____ faces.
- A square pyramid has _____ matching triangles.
- Use the _____ height, not the straight-up height, for the triangles.

WORKED EXAMPLE

fill in the blanks as we solve

A square pyramid has a base 10 cm on each side and a slant height of 7 cm. Find its surface area.

Step 1 Base: $10 \times 10 = \underline{\hspace{1cm}}$ cm².

Step 2 One triangle: $\frac{1}{2} \times 10 \times 7 = \underline{\hspace{1cm}}$ cm².

Step 3 Four triangles: $4 \times 35 = \underline{\hspace{1cm}}$ cm².

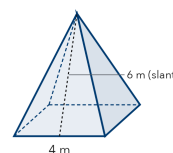
Answer SA = $100 + 140 = \underline{\hspace{1cm}}$ cm²

YOU TRY

use the example as a model

- Find the surface area of the square pyramid.

Show your work.



Answer _____ m²

- Find each area.

a. one triangle, base 6 in, slant 5 in _____ in² **b.** square base, side 6 in _____ in²

SUM IT UP

in your own words

To find the surface area of a pyramid, I add _____ and _____.

LESSON 5-8 • SESSION 2 • GUIDED NOTES

Lateral Area of Pyramids

Learning target – I can find the lateral area of a pyramid, which covers only its slanted faces. • 6.G.A.4

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
lateral area <i>área lateral</i>	The area of the slanted faces ____.	the base is left out
surface area <i>área superficial</i>	The area of ____ face, base included.	lateral area + base
lateral face <i>cara lateral</i>	A ____ side triangle of a pyramid.	3 on a triangle-base pyramid

KEY IDEA*complete each sentence*

- Lateral area = the area of all the ____ faces.
- Surface area = lateral area + ____ area.
- Read the problem to decide whether the base is ____.

WORKED EXAMPLE*fill in the blanks as we solve*

A square pyramid roof has a base 4 m on each side and a slant height of 6 m. Shingles cover only the slanted sides.

Step 1 Cover the 4 slanted triangles. Leave out the ____.

Step 2 One triangle: $\frac{1}{2} \times 4 \times 6 = \underline{\hspace{1cm}} \text{ m}^2$.

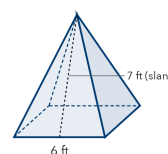
Step 3 Four triangles: $4 \times 12 = \underline{\hspace{1cm}} \text{ m}^2$.

Answer lateral area = $\underline{\hspace{1cm}} \text{ m}^2$

YOU TRY*use the example as a model*

1. Find the lateral area of the square pyramid.

Show your work.



Answer _____ ft^2

2. Find each lateral area.

a. SA = 90 m^2 , base = 25 m^2 _____ m^2

b. 3 faces, each 12 in^2 _____ in^2

SUM IT UP*in your own words*

I leave the base out when ____ because ____.
